Vedant Shinde

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linkedin.com/in/vedant-nilesh-shinde github.com/Vedant321 leetcode.com/u/Vedant_12345 Portfolio

Experience

Morgan Stanley *Oct 2021 – Jul 2025*
Senior Associate – Python Developer

- Engineered production **Python ETL pipelines** for IBM MQ transaction flows with **z-score anomaly detection**, enabling early failure signals that prevented **10+ monthly incidents** and supported automated infrastructure scaling decisions.
- Reduced decision-making time for stakeholders by **40%** across **15+ regulatory mandates** by designing a centralized **FastAPI data application** and **15+** analytical datasets tracking **100,000+ daily EOD transactions**, replacing manual reporting.
- Automated alert-parsing workflows by extracting asset signals from the data warehouse and auto-generating **Jira tickets** routed to **5** engineering squads, eliminating **10+** hours of manual daily ticket creation and reducing operational TOIL by **30%** for production managers.
- Architected a centralized **Prometheus and Grafana** telemetry pipeline ingesting logs from **50+ Java microservices** using **PromQL aggregations**, reducing **mean-time-to-diagnose by 10%** across production services.
- Optimized **Jenkins CI/CD pipelines** by removing redundant build steps and standardizing repository structure, reducing deployment effort by **10%** across **3** repositories.
- Led **15+ sprints**, conducted **5+ code reviews per week**, and mentored **2 engineers**, strengthening engineering standards, cross-team communication, and maintainability of Python & SQL data workflows.

Technical Skills

Programming: Python, SQL, R
Data Engineering: ETL/ELT, Data Pipelines, Apache Kafka, Apache Spark, PySpark, Spark Structured Streaming, Apache Airflow, dbt, Data Modeling, Dimensional Modeling, Star Schema, OLTP/OLAP, Data Warehousing, Data Quality, Hadoop
Databases: PostgreSQL, MySQL, Snowflake, Amazon Redshift, Databricks, Delta Lake, Unity Catalog, MongoDB
Cloud: AWS, Amazon S3, AWS Glue, Amazon EMR, Amazon Kinesis, Kinesis Data Firehose, AWS Lambda, IAM, Docker, CI/CD
Frameworks & Tools: Pandas, NumPy, Pytorch, FastAPI, Flask, Streamlit, Scikit-Learn, MLflow, Git, Jenkins, Linux

Projects

E-Commerce Lakehouse Analytics Platform (2026) |

- Built a production-scale **Databricks Lakehouse** processing **110M+ behavioral events** (**13.7GB**) from **5.3M users**, designing a **Bronze–Silver–Gold** architecture for scalable analytics and ML workloads.
- Engineered distributed **PySpark ETL pipelines** with Delta Lake, Unity Catalog, and Spark SQL for schema evolution, ACID transactions, partitioning, and incremental processing.
- Integrated **Apache Kafka and Spark Structured Streaming** to ingest and process continuous e-commerce events, extending the platform with near-real-time data processing alongside historical batch workloads.
- Created analytics-ready dimensional datasets through data validation, deduplication, sessionization, and business aggregations, supporting customer behavior, conversion funnels, revenue, and product analytics across **235K+ products**.

Supply Chain Data Pipeline (2026) |

- Implemented an end-to-end **data engineering pipeline** ingesting **600,000+ supply chain records** into PostgreSQL using modular Python, Pandas, and SQLAlchemy ingestion workflows.
- Modeled a normalized **3NF OLTP database** from raw CSV data, enforcing relational integrity and eliminating redundancy through SQL-based schema design.
- Created a **dbt transformation layer** to convert normalized operational data into an analytics-ready star schema, producing reusable fact and dimension models for supply chain analytics.
- Integrated automated **dbt data quality tests**, version-controlled SQL transformations, and deployed an interactive Streamlit dashboard on Render.

Open Source & Achievements

- **Open Source Contributor – AWS SDK for pandas:**
  Resolved nested decimal casting failures in **Athena Iceberg** writes by implementing recursive handling for **array, struct, and map** values, adding regression coverage, and validating the fix against a live AWS deployment.
- **1st Place – University at Buffalo AI for Good Hackathon (2026):**
  Won the **ACV Auctions challenge** by building **GoingOnce**, a multi-agent vehicle marketplace with **4 LangGraph agents** for vehicle search, history analysis, negotiation, and fleet distribution across **ACV Auctions and Copart**.
- **2nd Place – UCSF QBI Hackathon (2026):**
  Placed **2nd** by building **HiveBlot**, combining **OCR, computer vision, and LLMs** to convert Western blot figures from **15+ scientific papers** into a structured, searchable evidence index.

Education

University at Buffalo, SUNY *Aug 2025 – Dec 2026*
M.S. Data Science **GPA: 3.54 / 4.0**
Courses: Machine Learning, Statistical Learning, Probability, Data Models and Query Language, Data Warehousing, Reinforcement Learning

Vasantdada Patil Pratisthan's College of Engineering and Visual Arts *Jun 2017 – Jun 2021*
B.E. Information Technology